Curriculum Vitae

Personal Particulars

- Full Name: Emile Takahiro Okada
- Nationality: Norwegian

Employment History

• 2026-Current: Assistant Professor (Tenure-Track), Nanyang Technological University
• 2022-2026: Peng Tsu Ann Assistant Professor, National University of Singapore

Education

• 2018-2022: DPhil in Mathematics, University of Oxford
Thesis: The Wavefront Set of Representations of p-adic Groups
Supervisors: Prof. Dan Ciubotaru, Prof. Kevin McGerty

• 2017-2018: Master of Mathematics, University of Cambridge
Thesis: Modular Representations of Finite Groups of Lie Type in the Defining Characteristic.
Grade: Distinction.

• 2014-2017: Bachelor of Arts in Mathematics, University of Cambridge
Grade: First Class in all years.

Research

Authors listed alphabetically.

Preprints

9. Antor, J., Okada, E., *A Microlocal Description of Aubert-Zelevinsky Duality on Unipotent L-Parameters*

8. Karasiewicz, E., Okada, E., Wang, R., *The stable wave front set of the theta representation*, submitted

7. Gurevich, M., Okada, E., *Ramification of weak Arthur packets for p-adic groups*, submitted

Publications

6. Ciubotaru, D., Okada, E., *Local character expansions and asymptotic cones over finite fields*, Proceedings of the London Mathematical Society, 2025, Volume 130, Issue 5, doi.org/10.1112/plms.70046

5. Ciubotaru, D., Mason-Brown, L., Okada, E., *Wavefront Sets of Unipotent Representations of Reductive p-adic Groups II*, Journal fur die reine und angewandte Mathematik (Crelle's Journal), 2025, doi.org/10.1515/crelle-2025-0022

4. Ciubotaru, D., Mason-Brown, L., Okada, E., *Some Unipotent Arthur Packets for Reductive p-adic Groups*, International Mathematics Research Notices, 2024, Issue 9, Pages 7502-7525, doi.org/10.1093/imrn/rnad245

3. Ciubotaru, D., Mason-Brown, L., Okada, E., *The Wavefront Sets of Unipotent Supercuspidal Representations*, Algebra & Number Theory, Vol. 18 (2024), No. 10, 1863-1889, doi.org/10.2140/ant.2024.18.1863

2. Ciubotaru, D., Mason-Brown, L., Okada, E., *Wavefront Sets of Unipotent Representations of Reductive p-adic Groups I*, American Journal of Mathematics, 2025 (to appear, DOI: N/A)

1. Okada, E., *The Wavefront Set of Representations of p-adic Groups*, PhD thesis, University of Oxford

Successful Grant Applications

• *The Royal Society*: Newton International Fellowship
• *National University of Singapore, Institute for Mathematical Sciences*: for conference on wavefront sets in representation theory and number theory

Awards and Honors

• 2018-2022: Aker Scholarship
• 2014-2018: Cambridge University Trust Scholarship
• 2015, 2016, 2017, 2018: Emmanuel College Examination Prize

Invited Talks

• 26 invited talks at conferences and seminars at leading institutions including *Massachusetts Institute of Technology, University of Cambridge, University of Oxford, Tsinghua University, Peking University, Weizmann Institute of Science, Mathematisches Forschungsinstitut Oberwolfach.*

Conferences

• Mathematisches Forschungsinstitut Oberwolfach, Germany, December 2024, *Representations of p-adic groups*,
• Brin Mathematics Research Center, United States of America, October 2024, *Representation Theory and Lie Groups*,
• Vietnam Institute of Advanced Study in Mathematics, Vietnam, July 2024, *Pan Asian Number Theory Conference*,
• Australian National University, Australia, June 2024, *Pacific Rim Conference in Mathematics*
• Institute for Advanced Study in Mathematics, Hangzhou, China, November 2023, *Arthur Packets*
• Nisyros, Greece, July 2023, *Anaparastaseis: Orbits, Hecke algebras, and representations*
• National University of Singapore, Singapore, December 2022, *Representation Theory, Combinatorics and Geometry*
• Erwin Schrödinger International Institute for Mathematics and Physics, Austria, April 2022, *Minimal Representations and Theta Correspondence*,
• Durham University, Durham, UK September 2021, *Representations of p-adic groups and related topics*
• Centre International de Rencontres Mathématiques, Marseille, France, August 2019, *Buildings and Affine Grassmannians.*

Teaching

• *Lecturer - National University of Singapore*: MA5211 Lie Theory (x2), MA4263 Introduction to Analytic Number Theory, MA3201 Algebra II.
• *Tutor - Somerville College, University of Oxford*: Prelims Analysis II, Prelims Analysis III, Part A Integration, Part A Topology, and Part A Groups.
• *Teaching Assistant - Mathematical Institute, University of Oxford*: B2.1 Introduction to Representation Theory (x2), B2.2 Commutative Algebra (x1), B3.1 Galois Theory (x4), C2.2 Homological Algebra (x1), C2.7 Category Theory (x1).

Mentorship

• *PhD students*: Joven Teo Jia Xuan (unofficial) (2025 - current)
• *Master's students*: Timothy Peck Hsien Wei (2023 - 2024)
• *Undergraduate research project*: Galileo Grey (2024 - 2025), Malcolm Tan (2026)

Service

• *Journal Referee*: Journal für die reine und angewandte Mathematik, Selecta Mathematica, Advances in Mathematics, Algebra and Number Theory, International Mathematics Research Notices.
• *Seminar Organiser*: p-adic Arthur packets (2024), The local Langlands correspondence for unipotent representations (2022), Algebraic groups (2019).

Other Employment

• 01/2021 - 09/2021: Stipendiary Lecturer, *Somerville College, University of Oxford*
• 07/2018 - 08/2018: Foreign Exchange Desk Intern, *Den Norske Bank (DNB Bank ASA)*
• 06/2017 - 09/2017: Digitalization, Business Development and Innovation Intern, *Den Norske Bank (DNB Bank ASA)*, *Awards:* 1st place in hackathon.
• 06/2016-08/2016: Cambridge Image Analysis Group, *University of Cambridge*

References

• Prof. Dan Ciubotaru, *University of Oxford*, dan.ciubotaru@maths.ox.ac.uk
• Prof. Wee Teck Gan, *National University of Singapore*, matgwt@nus.edu.sg
• Prof. Ju-Lee Kim, *Massachusetts Institute of Technology*, juleekim@mit.edu
• Assoc. Prof. Maxim Gurevich, *Technion - Israel Institute of Technology*, maxg@technion.ac.il
• Asst. Prof. Lucas Mason-Brown, *The University of Texas at Austin*, lucas.masonbrown@gmail.com